

SQL INTERVIEW PATTERNS EVERY DATA ANALYST MUST KNOW

Learn the logic behind the most common SQL interview questions

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Why SQL Patterns Matter?

Instead of memorizing queries, interviews test pattern recognition.

If you know the pattern, you can solve any variation.

Commonly tested skills:

- Logical thinking
- Business understanding
- Clean query writing

In short: Think patterns, not syntax.

Aggregation + GROUP BY Pattern

Used when you see words like total, average, count, per category.

Example question: Find the average salary by department.

```
SELECT department, AVG(salary) AS avg_salary  
FROM employees  
GROUP BY department;
```

In short: GROUP BY summarizes data.

Top-N / Highest / Lowest Pattern

Used for top performers, highest sales, maximum values.

Example question: Find top 3 highest paid employees.

```
SELECT *  
FROM employees  
ORDER BY salary DESC  
LIMIT 3;
```

Your approach:

- Sort descending
- Limit results
- ORDER BY salary DESC LIMIT 3

The pattern: Sort first, then slice.

Duplicate Records Pattern

Used to identify duplicate emails, users, or records.

Example question: Find duplicate email IDs.

```
SELECT email, COUNT(*)  
FROM users  
GROUP BY email  
HAVING COUNT(*) > 1;
```

Your approach:

- Group by the field
- Count occurrences
- Filter where count > 1

JOIN Pattern

Used when data is spread across multiple tables.

Example question: Get customers and their orders.

```
SELECT c.name, o.order_id  
FROM customers c  
LEFT JOIN orders o  
ON c.id = o.customer_id;
```

Your approach:

- Identify the relationship
- Choose join type (LEFT keeps all customers)
- Match on foreign key

Missing Data / No Match Pattern

Used to find records without related data.

Example question: Customers who never placed an order.

```
SELECT c.*  
FROM customers c  
LEFT JOIN orders o  
ON c.id = o.customer_id  
WHERE o.order_id IS NULL;
```

Your approach:

- LEFT JOIN to orders
- Filter WHERE order_id IS NULL
- Those are your unmatched records

The pattern: NULL means "no match found".

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Window Function Pattern

Very common in analyst interviews. Used for ranking and comparisons.

Example question: Rank employees by salary.

```
SELECT name, salary,  
RANK() OVER (ORDER BY salary DESC) AS  
salary_rank  
FROM employees;
```

Your approach:

- Use RANK() or ROW_NUMBER()
- OVER() clause defines the window
- PARTITION BY for groups

The pattern: Window functions don't reduce rows like GROUP BY.

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Nth Highest Value Pattern

A classic interview question asked in many variations.

Example question: Find the 2nd highest salary.

```
SELECT MAX(salary)
FROM employees
WHERE salary < (SELECT MAX(salary) FROM
employees);
```

Your approach:

- Subquery for the max
- Find max below that value

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Date & Time Pattern

Used for daily, monthly, and yearly trend analysis.

Example question: Total orders per day.

```
SELECT DATE(order_date), COUNT(*)  
FROM orders  
GROUP BY DATE(order_date);
```

Your approach:

- Extract/format the date part
- GROUP BY that date
- COUNT or SUM metrics

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CASE WHEN Pattern

Used for applying conditional logic inside SQL queries.

Example question: Categorize employees by salary level.

```
SELECT name,  
CASE  
WHEN salary > 50000 THEN 'High'  
ELSE 'Low'  
END AS salary_level  
FROM employees;
```

In short: CASE works like IF–ELSE.

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Subquery Pattern

Used when filtering data based on aggregated results.

Example question: Employees earning above average salary.

```
SELECT *  
FROM employees  
WHERE salary > (SELECT AVG(salary) FROM  
employees);
```

In short: A query inside another query.

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CTE Pattern (WITH Clause)

CTEs improve readability, debugging, and logic flow.

Example question: Find the 2nd highest salary using CTE.

```
WITH ranked AS (  
SELECT *, DENSE_RANK() OVER (ORDER BY  
salary DESC) AS rnk  
FROM employees  
)  
SELECT * FROM ranked WHERE rnk = 2;
```

In short: CTE = clean, professional SQL.

Interview Tip

Interviewers focus more on:

- Your logic
- Pattern recognition
- How clearly you explain your approach

In short: Explain your thinking confidently.

Final Takeaway

If you understand these SQL patterns, you are interview-ready.

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